

Instructor: Dr. Jim D. Pham
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Office Hrs.: By Appointment
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Section: CS-113-03 (088989, TTH 6:30 PM, Fremont Campus, Rm 3 3404)

Course Description:

This course is an introduction to the discrete structures used in Computer Science with an emphasis on their applications. Topics covered include functions, relations, and sets; basic logic; proof techniques; basics of counting; graphs and trees; and discrete probability.

Prerequisites:

Advisory: CS 102 Introduction to Computer Programming Using C++

Textbook:

Discrete Mathematics with Applications, 5th Edition
Author: Susanna S. EPP
ISBN-13: 978-1-337-69419-3

Student Learning Outcomes

After completing the course, students will be able to:

1. Describe how formal tools of symbolic logic are used to model real-life situations, including those arising in computing contexts such as program correctness, database queries, and algorithms.
2. Relate the ideas of mathematical induction to recursion and recursively defined structures. Analyze a problem to create relevant recurrence equations.
3. Demonstrate different traversal methods for trees and graphs.
4. Apply the binomial theorem to independent events and Bayes' theorem to dependent events

Teaching Methods:

1. Lectures: The presentations slides, and assignments will be accessed through Ohlone Canvas. See Instructional Resources below.
2. Assignments: End-of-chapter activities and online activities will be assigned weekly to reinforce the material in the text. These assignments may require the application of various software packages.
3. Quizzes: Weekly unannounced quizzes may be given to help ensure students keep up with assigned material.
4. Participation: Student participation will be graded by the level of class participation on the Discussion Board. Each student will be required to do a minimum of 2 posts (1 post and 1 reply) per chapter on the Canvas Discussion Board.

Instructional Resources:

Canvas: Class documents, syllabus, lectures, assignments, and discussions will be posted on Canvas. You access Canvas at this URL: <https://ohlone.instructure.com>.

CANVAS CONFERENCES:

Depending on the complexity of the topics, I may conduct live lectures and labs using Canvas Conferences Zoom. The Zoom sessions will be posted on Canvas.

Email:

If you send me an email, then please put the **course number – name – topic** into the email **Subject** field. For example, CS 113 – Name – Issue

Discussion Board:

Your participation on the Discussion Board (i.e., Canvas) will be an important aspect of the class. Students will be required to post and respond to at least two posts on the discussion board.

Labs and Assignments:

Lab programming assignments will be assigned each week. It is the student's responsibility to submit the class assignments on the due date.

Exams and Tests:

There will be a) Lab Assignments; b) Online Participation; c) Quizzes; d) Midterm; e) Final Exam. You will have 1 attempt for each Chapter Exam and the Midterm and Final Exam. The final exam will not be comprehensive in nature. However, the instructor reserves the right to retest material that was not appropriately comprehended. The questions will be programming, multiple-choice, true/false, and short-answer fill-ins. All exams will be timed, however.

Policy on Student Work:

Labs and assignments are to be done individually unless a group project is specified. You are encouraged to talk to fellow students about concepts and methods, but you may not copy someone else's work. I adhere to the **Ohlone College Policy on Academic Integrity**.

GRADING:

Canvas Participation:	10%
Quizzes:	10%
Labs:	30%
Midterm:	20%
Final Exam:	30%

GRADING SCALE:

A = 90-100
B = 80-89
C = 65-79
D = 55-64
F = <55

Course Schedule: The following is a recommended schedule. The instructor reserves the right to make any schedule changes deemed necessary.

Week	Date	Activity
1	01/24/2023	Chapter 1 Speaking Mathematically - Variables
	01/26/2023	- Sets
2	01/31/2023	- Relations and Functions - Lab
	02/02/2023	- Graphs
3	02/07/2023	Chapter 2 The Logic of Compound Statements - Logical Form and Logical Equivalence - Lab
	02/09/2023	- Conditional Statements
4	02/14/2023	- Conditional Statements (cont.) Holiday: President's Day Friday, February 18 – Monday, February 21 (weekday and weekend classes do not meet; Ohlone College Closed)
	02/15/2023	- Valid and Invalid Arguments
5	02/21/2023	- Application: Digital Logic Circuits
	02/23/2023	- Application: Number Systems and Circuits for Addition
6	02/28/2023	Chapter 3 The Logic of Quantified Statements - Predicates and Quantified Statements I - Lab
	03/02/2023	- Predicates and Quantified Statements II
7	03/07/2023	- Statements with Multiple Quantifiers
	03/09/2023	- Arguments with Quantifiers Statements
8	03/14/2023	Chapter 4 Elementary Number Theory and Methods of Proof - Direct Proof and Counterexample 1: Introduction - Lab
	03/16/2023	- Direct Proof and Counterexample 2: Writing Advice
	03/20/2023 03/26/2023	Spring Break Monday, March 20 – Sunday, March 26 (Weekday and weekend classes do not meet)
9	03/28/2023	- Direct Proof and Counterexample 3: Rational Numbers and Divisibility
	03/30/2023	- Indirect Argument: Contradiction and Contraposition - Two Classical Theorems
10	04/04/2023	Midterm
	04/06/2023	Chapter 5 Sequences, Mathematical Induction, and Recursion - Lab

11	04/11/2023	Chapter 6 Set Theory - Lab
	04/13/2023	- Chapter 6 continuing
12	04/18/2023	Chapter 7 Functions - Lab
	04/20/2023	- Chapter 7 continuing
13	04/25/2023	Chapter 8 Relations - Lab
	04/27/2023	- Chapter 8 continuing
14	05/02/2023	Chapter 9 Counting and Probability - Lab
	05/04/2023	- Chapter 9 continuing
15	05/09/2023	Chapter 10 Graphs and Trees - Lab
	05/11/2023	Final Exam Review
16	05/16/2023	Final Exam